

1. Abstract

This paper compares the forecasting performance of an autoregressive integrated moving average model with exogenous regressors (ARIMAX) with two architectures of deep-learning neural networks: a feed-forward neural network (FFNN) and a recurrent neural network (RNN) of the type Gated Recurrent Unit (GRU), as well as with a naïve benchmark: the no-change model. The forecast variable is the quarterly unemployment rate for a panel of 22 OECD economies from 1994 to 2019, whereas the two other features for neural networks and two exogenous regressors for the ARIMAX are the quarterly real gross domestic product and the quarterly CPI inflation rate — their use is motivated by the potential existence of the relationships with the unemployment rate, namely the Okun's Law for the former and the Phillips Curve for the latter. The hyperparameter tuning was carried out by using grid search for hyperparameters and evaluating the performance of each combination on a validation set, choosing the best-performing one for subsequent forecasting. Both the neural network architectures achieved the lowest validation loss using a 4-year window of past observations as inputs, consistent with the hysteresis theory of unemployment persistence by Blanchard and Summers. The performance of each model was then compared using a rolling-origin expanding-window forecasting evaluation for each model. The RNN achieved the lowest MAE and RMSE, followed closely by ARIMAX, and both outperformed the naïve benchmark significantly at a 5% significance level; the FFNN demonstrated the poorest performance. An additional robustness test was performed using the 2020-2023 period of the COVID recession as the test data for rolling-origin evaluation, using fixed hyperparameters from the main pre-pandemic test period, to evaluate the performance of the forecasting models when confronted with a substantial economic shock. The results showed a different dynamic: although RNN's forecasts had slightly lower MAE and RMSE, neither model was able to produce better forecasts than the naïve method, while ARIMAX was significantly worse compared to it in the Diebold-Mariano test. Furthermore, the examination of the ARIMAX's exogenous regressors' coefficients shows strengthening of the Okun's Law relationship during the pandemic.

2. Introduction

In the present era of rapid advances in machine learning and, more specifically, deep learning neural networks, aside from their use for more conventional purposes such as natural language processing and computer vision, their use for forecasting has also accelerated. In econometrics, ARIMA models have long been the staple for modelling and forecasting macroeconomic indicators. This paper explores and compares the performance of an ARIMAX model with deep-learning neural networks in forecasting the unemployment rate. The latter is an important macroeconomic variable, as it measures the overall well-being of an economy and its workforce. The ability to forecast it to a high degree of accuracy is immensely important to national governments, because they would be able to act on any tendencies early; for example, by managing the aggregate demand in accordance with the principles of Keynesianism, or implementing supply-side policies to ensure a stable employment outlook.

The data and findings cover a panel of 22 OECD economies. Aside from the time series for the unemployment rate itself used as input for the models, RGDP growth and CPI inflation were also used as exogenous regressors for ARIMAX and as additional input features in neural networks. According to Okun's law, RGDP growth is negatively correlated with the unemployment rate, as the labour markets tighten during higher growth due to more hiring

for higher output; and vice versa, labour markets loosen when growth slows, as more job seekers compete for a lower number of vacancies when hiring slows. Furthermore, according to the concept of the Phillips curve, in the short run, the unemployment rate is negatively correlated with the inflation rate: as the labour markets tighten, workers attain more bargaining power, and available vacancies offer higher wages to compete for scarce labour, therefore increasing disposable income and consumption; by contrast, as labour markets loosen, the effect of bargaining power and higher wages is diminished, leading to a decrease in consumption and inflation rate.

The advantage of deep-learning neural networks is that they can exploit the non-linear patterns in the input data, as well as be trained on the data for the entire panel of countries at once, allowing for exploitation of cross-country patterns as well. On the other hand, ARIMAX models need to be fitted per country, and can only work with linear relationships. However, due to their regression nature, they are far more suitable for modelling data, whereas neural networks are nothing short of black boxes. Furthermore, it is far easier computationally to fit an ARIMAX model than a neural network; some methodology decisions were therefore influenced by the available computational resources.

3. Data

Data used in this project were obtained from the official OECD Data Explorer.¹ Each variable encompasses a time series from 1993 to 2019 for a panel of 22 OECD member states: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Ireland, Italy, Japan, Korea, Luxembourg, Mexico, the Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, the United Kingdom, and the United States. The time series contain no gaps for each variable.

3.1. Target variable

The target variable is the seasonally adjusted quarterly unemployment rate for the economically active population aged 15 or older; the quarterly values were obtained by averaging the monthly unemployment rate across all months in a given quarter.

3.2. Explanatory variables

The first explanatory variable is the seasonally adjusted real gross domestic product quarter-on-quarter growth expressed as a percentage.

The second explanatory variable is the year-on-year CPI inflation, which was obtained from unadjusted raw CPI indexes and expressed as a percentage. Because of the need to ensure that the data for the models have no gaps for each variable, only the values starting from Q1 1994 were used, as it was impossible to compute year-on-year CPI inflation for 1993.

3.3. Preprocessing

To ensure stationarity of the time series for all the variables, a combination of Augmented Dickey-Fuller and Kwiatkowski-Phillips-Schmidt-Shin tests was used. It was determined that the year-on-year CPI inflation is not stationary for 6 countries in the panel, and the unemployment rate is not stationary for 9 countries in the panel. For a fair comparison between the models, CPI inflation was differenced uniformly before being used further in the models; the unemployment rate itself was differenced as part of the neural networks' inputs, whereas for ARIMAX, the models using an I order of either 0 or 1 were tested for each country at the tuning stage (Hewamalage et al., 2022). RGDP growth was stationary for all the countries of the panel; therefore, it was left undifferenced. The inputs for neural networks

¹ <https://data-explorer.oecd.org/>

were also normalised, as normalisation is known to improve the speed and efficiency of gradient descent (Chollet and Watson, 2025). After producing the forecasts, their values were subsequently un-normalised and un-differenced.

3.4. Data split

For forecasting, the data were divided into three periods: initial training (1994-2009, inclusive), validation set (2010-2014, inclusive), and testing (2015-2019, inclusive). To test the performance of models during the COVID recession as a separate test, data for the same panel, except for Japan, were used with the following split: training (1994-2019, inclusive), testing (2020-2023, inclusive); Japan was excluded due to its unemployment data discontinuity past 2020.

4. Methodology

4.1 Naïve benchmark

The naïve benchmark in this research is the no-change model, where the last known observation is used as a forecast without any changes

4.2. Loss functions and error computation

The loss function that was minimised for both neural network models during training is the Mean Squared Error (MSE). For model evaluation, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were used, as they preserve the units, and MAE specifically is the most easily interpretable in the context of this paper; MAPE was discarded for both use cases, as, firstly, the variable being forecast is already a percentage, and secondly, it was likely to produce excessively inflated errors given the near-zero values of differenced series (Hewamalage et al., 2022).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

4.3. Rolling-origin expanding-window evaluation

Rolling-origin evaluation over the test period was chosen over a fixed-origin design, as it produces forecasts for one time step ahead for each origin, which allows for a greater number of forecasted values to test the performance of each model, and because it avoids potential errors driven by the influence of occurrences unique to a fixed origin, data for which may be different from the historical outlook (Tashman, 2000). Furthermore, as the time series are relatively short, all models used rolling-origin, expanding-window evaluation over the test period to produce the forecasts (Hewamalage et al., 2022). After selecting the parameters of a model, the validation set was appended to the initial training set to form the initial training set for rolling-origin expanding-window forecasting evaluation. At every time

step in the training data, the model was refitted using the training data and produced a forecast value for the current origin per country; the true values for that origin were then appended to the training data and used to fit the model to forecast the next origin's values.

4.4. Feed-forward Neural Network (FFNN)

The first model is the deep-learning feed-forward neural network that takes previous values of both endogenous and exogenous variables as features; uses MSE as its loss function and Rectified Linear Unit (ReLU) as the activation function in the hidden layers. During the inputs/targets generation process, windows of N timesteps were generated per country, with the corresponding target being the value of the endogenous variable in the next time step after the last one in the input window. As an FFNN is unable to work with temporally structured data, unlike an RNN, the input window was flattened, and each individual sample therefore became a vector of shape $(\text{num_features} * N)$. The training was performed over the entire panel of countries to take advantage of potential cross-country relationships. The hyperparameters' selection was performed using a grid search. For each combination of hyperparameters, the model was trained on the initial training data for the entire panel and validated on the validation set. To reduce computational costs, rolling-origin forecasting evaluation was not used during hyperparameter tuning, and an early stopping callback was implemented for each individual model in the grid search, which halted training after 10 epochs if the validation loss had not decreased by more than 0.005 units (Chollet and Watson, 2025). The model that achieved the best validation loss has the following structure: the first hidden layer with 32 neurons, followed by the second hidden layer with 8 neurons, followed by a dropout layer with a dropout rate of 0.2, and then followed by a linear layer to produce the value of the forecast. For this model, the best validation loss was achieved after 40 epochs, using a batch size of 16 and values for 16 past quarters as inputs, or the past 4 years before the time step for which forecasting is taking place. After selecting the hyperparameters, rolling-origin expanding-window evaluation was performed over the test period.

4.5. Recurrent Neural Network (RNN)

The second model is the deep-learning recurrent neural network, more specifically the Gated Recurrent Unit (GRU), that takes previous values of both endogenous and exogenous variables as features and uses MSE as its loss function. As with the FFNN, during the inputs/targets generation process, windows of N timesteps were generated per country, with the corresponding target being the value of the endogenous variable in the next time step after the last one in the input window. However, RNNs are able to work with temporally structured data, so the input window was not flattened, and each individual sample therefore became a vector of shape $(N, \text{num_features})$. The training was performed over the entire panel of countries to take advantage of potential cross-country relationships. The hyperparameters' selection was performed using a grid search, using the same principle as the FFNN, with the early stopping callback with the same parameters used. The model that achieved the best validation loss has the following structure: the first GRU layer with 32 units and a dropout rate of 0.2, followed by the second GRU layer with 16 units and a dropout rate of 0.2, and then followed by a linear layer to produce the value of the forecast. The best validation loss was achieved after 8 epochs, using a batch size of 64 and values for 16 past quarters as inputs, or the past 4 years before the time step for which forecasting is taking place. After selecting the hyperparameters, rolling-origin expanding-window evaluation was performed over the test period.

4.6. Exogenous ARIMA (ARIMAX)

The last model is the exogenous autoregressive moving average (ARIMAX). The non-seasonal version was chosen due to seasonal patterns already being eliminated in the data, as the RGDP growth and the unemployment rate were already seasonally adjusted, and the CPI inflation was year-on-year. Unlike deep-learning neural networks, ARIMAX can only be fitted per country. The exogenous variables were RGDP growth and year-on-year CPI inflation, whereas the endogenous variable was the unemployment rate. The values of the exogenous variables were shifted by one time step relative to the endogenous variable so that, for every forecast origin, the model could use the exogenous variables from the previous time step and forecast the value of the endogenous variable. For computational efficiency, the orders of the per-country models were first estimated by running an Auto ARIMAX that minimised the Akaike Information Criterion (AIC). After that, for each country, a grid search of the best order was performed in the neighbourhood of 2 more or less AR and MA orders than the ones suggested by the Auto ARIMAX, as well as testing both 0 and 1 for the I order in combination with each AR/MA pair. For each order, the model was evaluated using rolling-origin expanding-window forecasting over the validation data while refitting at every origin, and the MSE of forecasts was recorded per model. The fitting was achieved using gradient-based optimisation, and if the model's weights did not converge, then fitting using gradient-free optimisation was attempted with a larger number of iterations. After tuning the models, rolling-origin expanding-window evaluation was performed over the test period using the best models per country. At each origin, the models' weights were warm-started from the previous origin's converged parameters.

4.7. Models' performance evaluation

Firstly, the performance of models was evaluated by comparing the MAE and RMSE of their forecasts. Secondly, the Diebold-Mariano test was run over the combinations of forecasts' MAE of two models averaged across countries per time step: the loss differentials were computed, averaged, and standardised to evaluate the significance at the 5% significance level (Hewamalage et al. 2022).

4.8. COVID recession performance evaluation

The performance of the models during the COVID recession has also been tested, when all three macroeconomic variables being inspected diverged considerably from their pre-recession levels. With the exception of naïve forecast, which doesn't need any change, hyperparameters selected in the previous test were saved and used for this test without any change; rolling-origin expanding-window evaluation was performed over the test period. This evaluates the generalisation of the models trained on pre-recession data to the volatility brought by it, rather than re-optimising them on new data immediately.

5. Results

5.1. 2015-2019 test period

Table A shows the MAE and RMSE of forecasts per model over the 2015-2019 period aggregated across the full 22-country panel. All values rounded to 4sf

Table A

Model	MAE	RMSE
Naïve	0.1981	0.2756
FFNN	0.1823	0.2543
RNN	0.1742	0.2392
ARIMAX	0.1773	0.2450

Table B shows the results of Diebold-Mariano tests for the forecasts by each pair of models over the 2015-2019 period; the forecasts' MAE were averaged across countries for each time step. All values rounded to 3sf

Table B

Model A	Model B	D-M statistic	p-value	Significance at 5%	Preferred model
Naïve	FFNN	1.87	0.062	No	FFNN
Naïve	RNN	4.051	<0.001	Yes	RNN
Naïve	ARIMAX	2.726	0.006	Yes	ARIMAX
FFNN	RNN	1.483	0.138	No	RNN
FFNN	ARIMAX	0.785	0.433	No	ARIMAX
RNN	ARIMAX	-0.625	0.532	No	RNN

*Preferred model reflects the sign of the Diebold-Mariano mean error differential only; only differences marked significant at 5% have shown statistically significantly better performance

5.2. 2020-2023 COVID test period

Table C shows the MAE and RMSE of forecasts per model over the 2020-2023 period aggregated across the full 21-country panel. All values rounded to 4sf

Table C

Model	MAE	RMSE
Naïve	0.4251	0.8648
FFNN	0.5882	1.536
RNN	0.3949	0.8459
ARIMAX	0.5246	1.1365

Table D shows the results of Diebold-Mariano tests for the forecasts by each pair of models over the 2020-2023 period; the forecasts' MAE were averaged across countries for each time step. All values rounded to 3sf

Table D

Model A	Model B	D-M statistic	p-value	Significance at 5%	Preferred model
Naïve	FFNN	-1.445	0.149	No	Naïve
Naïve	RNN	1.5	0.134	No	RNN
Naïve	ARIMAX	-2.349	0.019	Yes	Naïve
FFNN	RNN	1.681	0.093	No	RNN
FFNN	ARIMAX	0.714	0.475	No	ARIMAX
RNN	ARIMAX	-2.748	0.006	Yes	RNN

*Preferred model reflects the sign of the Diebold-Mariano mean error differential only; only differences marked significant at 5% have shown statistically significantly better performance

6. Discussion

6.1. 2015-2019 test period

During the main test where years 2015 to 2019 were included in the rolling-origin evaluation, RNN showed the strongest MAE and RMSE, followed closely by ARIMAX, and both of them beat the naïve model significantly at the 5% significance level in the Diebold-Mariano test. FFNN showed the worst performance, being the worst of all models in MAE and RMSE, though not significantly worse than any of them in the Diebold-Mariano test.

During the tuning stage, both the RNN and FFNN showed the best performance on the validation data with exactly 16 past quarters of inputs, despite both higher and lower values being included in the grid search. This suggests that unemployment has significant persistence, with information as far as 4 years back contributing to its prediction. This finding is consistent with Blanchard and Summers' hysteresis theory, which challenges the notion that the unemployment rate returns to a fixed equilibrium rate following the shock (Blanchard and Summers, 1986). If this were the case, the models would have likely performed better with fewer past lags as inputs, as increasing the number of those is likely to increase the noise in the data. The finding that information from a relatively long historical window improves forecasting performance is therefore consistent with the idea that past unemployment contains information about the current and future unemployment rate, beyond the direct effects of a shock, and that the rate may not return to the previous NAIRU.

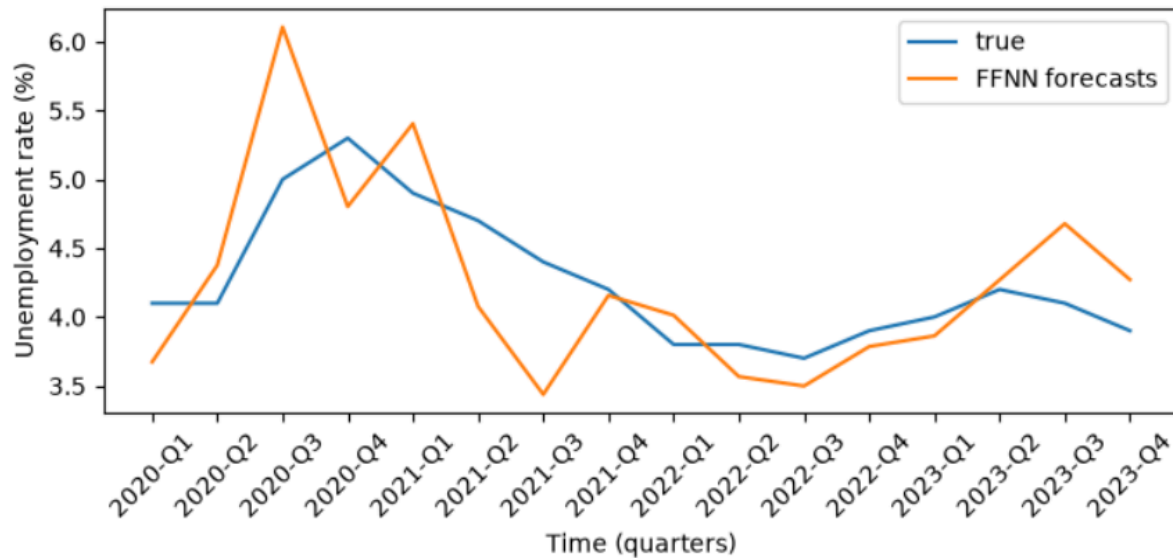
6.2. 2020-2023 COVID test period

During the additional COVID test, the forecast errors of the models show a substantial divergence compared to the main test period of 2015-2019. All models performed worse due to the shock, and only the RNN demonstrated better MAE and RMSE than the naïve model, though the former could not produce significantly better forecasts at the 5% significance level in the Diebold-Mariano test compared to the latter. Both the FFNN and ARIMAX, on the other hand, demonstrated higher MAE and RMSE than the naïve model, though only ARIMAX was significantly worse than the naïve benchmark in the Diebold-Mariano test. This is a notable change: none of the models could significantly outperform the naïve method in the Diebold-Mariano test once the test data included a major economic crisis, despite the fact that all the architectures were still being re-fitted at every origin.

6.3. Examination of model forecasts for the UK economy over the 2020-2023 COVID test period

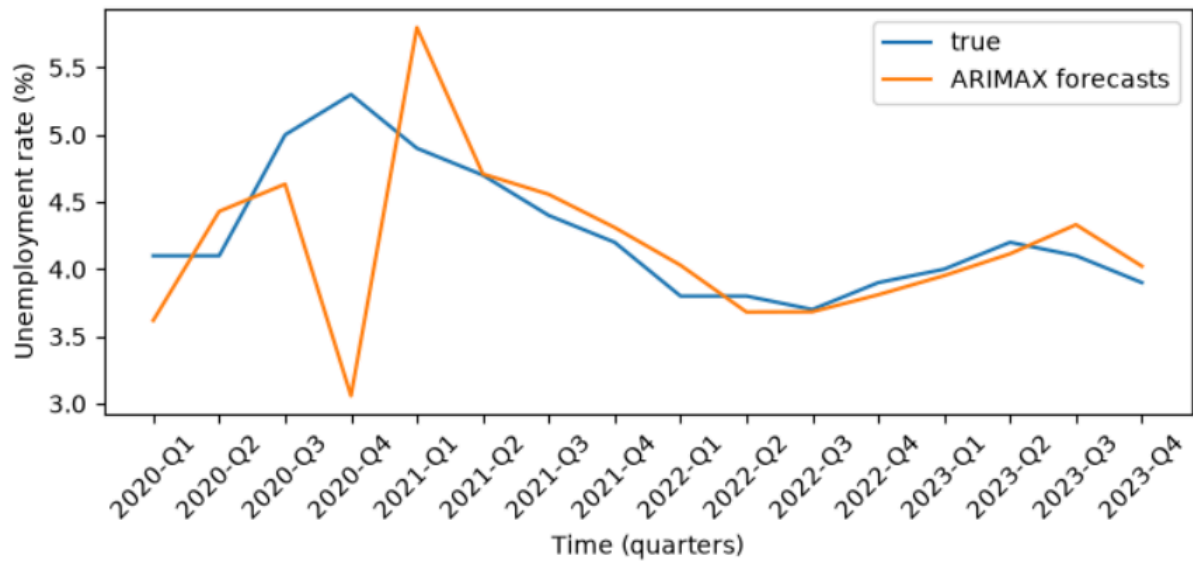
As the plot of FFNN rolling-origin forecasts for the UK in Figure 1 shows, the model was able to correctly identify the recession upward trend at first, but its later forecasts of the downward trend kept overestimating the intensity of the decrease, both at 2020 Q4 and 2021 Q2 and Q3. Furthermore, after following the true values more closely from 2021 Q4 to 2023 Q3, it overestimated the upward trend in 2023 Q3, as if the model expected another recession-driven surge in the unemployment rate to start.

Figure 1



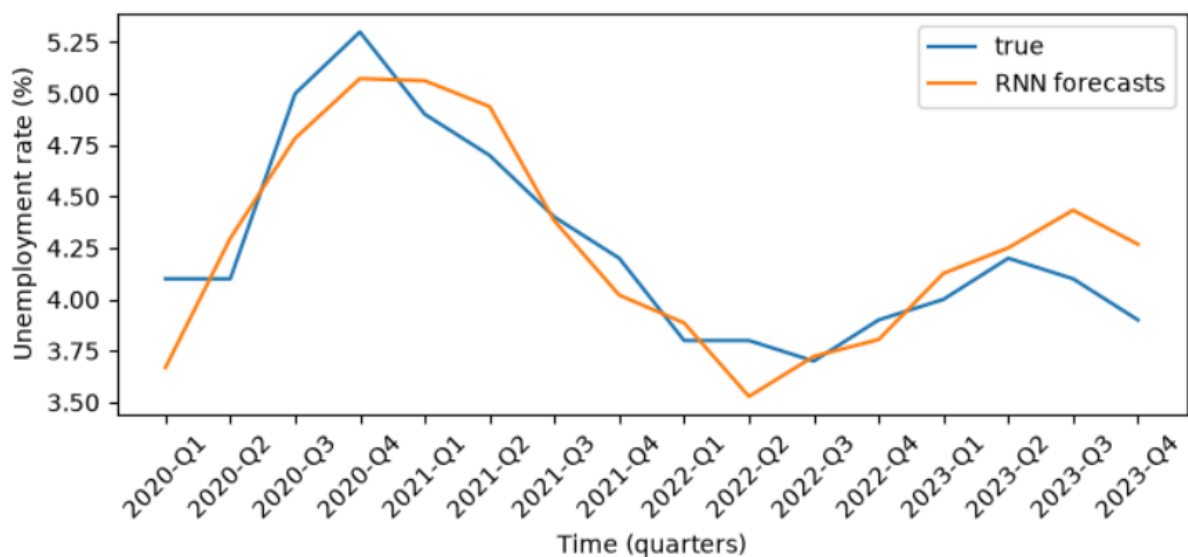
ARIMAX showed an even worse performance than the FFNN during the 2020 Q1 to 2021 Q1 period, as it first forecast a substantial drop in the UK unemployment rate in 2020 Q4, and then overshoot the actual value in 2021 Q1, while the variable was decreasing steadily in reality in Figure 2. This reflects the chosen ARIMAX order for the UK that showed the least MSE during the fitting stage over the 2010-2014 validation period: (1, 1, 0). The model only has knowledge of one past true value before making a forecast, which explains the erratic forecasts' behaviour. An attempt was made to level the field by including the AR and MA orders of 16 in the grid search; however, the models with those parameters were numerically unstable. Furthermore, another limitation might be that the time series is still not stationary after one differencing; the mean and variance are likely to have increased significantly over the recession period. From 2021 Q2, however, as the volatility of the unemployment rate has decreased, the forecasts followed the true values quite closely, which explains why ARIMAX was able to achieve lower MAE and RMSE than FFNN, as well as be significantly better than the latter in the Diebold-Mariano test.

Figure 2



RNN showed the best performance compared to all the other models tested, showing significantly better performance in the Diebold-Mariano test. Judging by its performance in forecasting the UK unemployment rate in Figure 3, it tended to underestimate the initial surge, while overestimating the values in 2021 Q1 and Q2. The forecast performance from 2023 Q3 onwards was worse than that of ARIMAX, yet the RNN was still able to produce significantly better forecasts.

Figure 3



6.4. ARIMAX exogenous regressors examination

The summaries of exogenous regressors' coefficients for ARIMAX were computed per country for both the 2015-2019 test period and the 2020-2023 test period to examine the

implied Okun's law and Phillips curve relationship. A comparison of the pre-COVID and COVID period showed a substantial strengthening of the RGDP growth relationship with the unemployment rate during COVID. Pre-COVID, negative and majority-significant coefficients were observed in only 2 of 22 countries (Korea, Mexico); during COVID, this was observed in 8 of 21 countries (Belgium, France, Italy, Korea, Mexico, Norway, Portugal, United States). On the other hand, the COVID period CPI inflation results show the same pattern as pre-COVID — negative sign in most countries but significant only in several: Denmark, Ireland, New Zealand. Mexico, on the other hand, has a coefficient that is consistently positive and significant in both periods, which suggests that its inflation is likely driven by other factors than the influence of labour markets. Unfortunately, it is impossible to compute the equivalent metric for deep-learning neural networks, as the input data is transformed non-linearly to generate an input.

7. Limitations

7.1. Model architecture limitations

Firstly, a limitation of my research is that I do not have sufficient compute resources to compare a larger number of models in the grid search. Ideally, the performance of each FFNN and RNN config from the grid should have been evaluated by running a rolling-origin forecast over the validation test set, as it was done with ARIMAX. However, that would require vastly more compute and time. Furthermore, the neural networks' configs that produced the least validation loss on the validation data set are not guaranteed to produce the best forecasts during the rolling-origin evaluation.

Secondly, the idea to let the deep-learning neural networks exploit cross-country patterns by training them on the data for the whole panel of countries at once means that they do not get a separate model per country, which might have been able to perform better. ARIMAX, on the other hand, can work with only one country at a time; although it was an option to add the entire panel as exogenous regressors, this was not implemented, as the impact of adding so many exogenous variables would have likely been either negligible or negative.

7.2. Data limitations

The panel of countries was chosen based on data homogeneity and stable availability for all economies. A number of countries were excluded, as they did not have stable coverage for all the metrics. Therefore, the size of the data set used is relatively small for the neural networks; measures such as neuron dropout were added to mitigate overfitting, but FFNN and RNN are still likely to perform worse on countries not included in the original panel.

The only pre-processing included differencing and data normalisation for neural networks; however, the heteroscedasticity was not accounted for in any way.

Moreover, the OECD data is likely to have been revised and adjusted post-factum. This means that models are trained on final, revised data rather than the provisional figures that would have been used when forecasting actual future values. Therefore, the reported accuracy may be somewhat optimistic compared to realistic forecasting conditions.

7.3. Evaluation limitations

The Diebold-Mariano test used is the simplest version and does not account for the small sample size, which ideally requires the Harvey-Leybourne-Newbold correction; this may mean that some of the test results are false positives. Furthermore, averaging the error differentials across the panel of countries was performed to produce a unified metric for the entire panel. This may mask the overall poor performance in some countries, as this might have been averaged out while comparing the models.

8. Conclusion

This paper compared the forecasting performance of ARIMAX, FFNN and GRU models against a naïve benchmark for the quarterly unemployment rate across a panel of 22 OECD economies, using RGDP growth and CPI inflation as exogenous regressors, motivated by Okun's Law and the Phillips curve. Over the pre-COVID 2015-2019 test period, the RNN achieved the lowest MAE and RMSE overall, closely followed by ARIMAX, and both significantly outperformed the naïve benchmark at the 5% level; the FFNN was unable to do so despite outperforming the naïve model on raw error alone. The finding that both neural network architectures achieved their best validation loss when using a 4-year input window is consistent with the hysteresis theory of unemployment persistence proposed by Blanchard and Summers (1986), suggesting that the unemployment rate does not simply revert to the pre-shock fixed equilibrium.

The COVID-19 robustness test, however, showed considerably different results: none of the models significantly beat the naïve benchmark, and ARIMAX in particular performed significantly worse than the no-change model despite being refitted at every origin. This is consistent with the examination of ARIMAX's exogenous coefficients, which showed a substantial strengthening of the Okun's Law relationship specifically during the pandemic-caused recession, which suggests that the linear structure ARIMAX relies on became inefficient once the underlying relationship shifted, whereas the RNN's relatively stable performance points to some advantage in modelling that shift non-linearly. Overall, my results suggest that a linear model such as ARIMAX, while a strong performer under stable conditions, is likely to generalise less reliably than a regularised recurrent neural network once a major economic shock disrupts the relationships the model depends upon.

Future research can apply the Harvey-Leybourne-Newbold correction to the Diebold-Mariano tests, evaluate an LSTM architecture alongside the GRU, or fit the neural networks per-country to assess whether the cross-country training approach used here helped or hindered performance during the pandemic shock.

9. Reference list

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10. Appendix

10.1. MAE and RMSE per model and per country over the 2015-2019 test period

Model	Country	MAE	RMSE
Naive	Australia	0.1117	0.1411
Naive	Austria	0.1983	0.2403
Naive	Belgium	0.315	0.3723
Naive	Canada	0.1233	0.1483
Naive	Denmark	0.185	0.2311
Naive	Finland	0.245	0.3725
Naive	France	0.1667	0.2025
Naive	Germany	0.0917	0.0986
Naive	Ireland	0.32	0.391
Naive	Italy	0.225	0.2905
Naive	Japan	0.0717	0.0916
Naive	Korea	0.1233	0.1609
Naive	Luxembourg	0.11	0.1563
Naive	Mexico	0.1272	0.1404
Naive	Netherlands	0.2017	0.2303
Naive	New Zealand	0.2	0.2757
Naive	Norway	0.2133	0.2558
Naive	Portugal	0.4083	0.5175
Naive	Spain	0.495	0.5556
Naive	Sweden	0.1667	0.2383
Naive	United Kingdom	0.14	0.1581
Naive	United States	0.1183	0.1443
FFNN	Australia	0.1242	0.1473
FFNN	Austria	0.2016	0.2551
FFNN	Belgium	0.3088	0.3507
FFNN	Canada	0.1012	0.1216
FFNN	Denmark	0.17	0.2192
FFNN	Finland	0.2513	0.3509
FFNN	France	0.1355	0.1784
FFNN	Germany	0.0723	0.085
FFNN	Ireland	0.3882	0.5189
FFNN	Italy	0.2506	0.2992
FFNN	Japan	0.0767	0.0904

FFNN	Korea	0.1436	0.1948
FFNN	Luxembourg	0.1408	0.1808
FFNN	Mexico	0.143	0.1646
FFNN	Netherlands	0.1191	0.1478
FFNN	New Zealand	0.1966	0.2751
FFNN	Norway	0.2061	0.2552
FFNN	Portugal	0.3223	0.4077
FFNN	Spain	0.2693	0.323
FFNN	Sweden	0.176	0.231
FFNN	United Kingdom	0.1233	0.1505
FFNN	United States	0.0899	0.1184
RNN	Australia	0.1144	0.1407
RNN	Austria	0.1927	0.2397
RNN	Belgium	0.3167	0.3713
RNN	Canada	0.1122	0.1302
RNN	Denmark	0.1824	0.2401
RNN	Finland	0.2461	0.3583
RNN	France	0.1567	0.1983
RNN	Germany	0.0648	0.0768
RNN	Ireland	0.2376	0.3183
RNN	Italy	0.2377	0.2824
RNN	Japan	0.064	0.0866
RNN	Korea	0.1338	0.1767
RNN	Luxembourg	0.1437	0.179
RNN	Mexico	0.1178	0.137
RNN	Netherlands	0.1289	0.1484
RNN	New Zealand	0.2079	0.2937
RNN	Norway	0.2095	0.2599
RNN	Portugal	0.3089	0.3997
RNN	Spain	0.2572	0.3026
RNN	Sweden	0.1889	0.2524
RNN	United Kingdom	0.1199	0.1414
RNN	United States	0.0912	0.1168
ARIMAX	Australia	0.1166	0.159
ARIMAX	Austria	0.1959	0.2362
ARIMAX	Belgium	0.3409	0.4024

ARIMAX	Canada	0.1533	0.1757
ARIMAX	Denmark	0.1874	0.2262
ARIMAX	Finland	0.228	0.3625
ARIMAX	France	0.1638	0.2155
ARIMAX	Germany	0.0573	0.0653
ARIMAX	Ireland	0.2161	0.2707
ARIMAX	Italy	0.2443	0.2976
ARIMAX	Japan	0.0676	0.0824
ARIMAX	Korea	0.1264	0.1686
ARIMAX	Luxembourg	0.1267	0.1667
ARIMAX	Mexico	0.1266	0.1422
ARIMAX	Netherlands	0.1378	0.1689
ARIMAX	New Zealand	0.2195	0.2829
ARIMAX	Norway	0.2098	0.2528
ARIMAX	Portugal	0.2991	0.4381
ARIMAX	Spain	0.2192	0.2686
ARIMAX	Sweden	0.2159	0.2957
ARIMAX	United Kingdom	0.138	0.1657
ARIMAX	United States	0.1107	0.1286

10.2. MAE and RMSE per model and per country over the 2020-2023 COVID test period

Model	Country	MAE	RMSE
Naive	Australia	0.3712	0.5703
Naive	Austria	0.5021	0.7483
Naive	Belgium	0.3583	0.4604
Naive	Canada	1.0625	2.0119
Naive	Denmark	0.3792	0.5302
Naive	Finland	0.3896	0.5139
Naive	France	0.3229	0.5034
Naive	Germany	0.1271	0.1648
Naive	Ireland	0.5167	0.7471
Naive	Italy	0.3917	0.5459
Naive	Korea	0.1729	0.2457
Naive	Luxembourg	0.3896	0.5245
Naive	Mexico	0.2371	0.3915

Naive	Netherlands	0.3042	0.3621
Naive	New Zealand	0.2563	0.3865
Naive	Norway	0.375	0.4634
Naive	Portugal	0.4354	0.5896
Naive	Spain	0.4792	0.6606
Naive	Sweden	0.4063	0.5038
Naive	United Kingdom	0.25	0.326
Naive	United States	1.2	2.5979
FFNN	Australia	0.5591	1.131
FFNN	Austria	0.772	1.586
FFNN	Belgium	0.33	0.4141
FFNN	Canada	1.9201	3.9857
FFNN	Denmark	0.4481	0.5773
FFNN	Finland	0.4634	0.6169
FFNN	France	0.3108	0.456
FFNN	Germany	0.2526	0.3365
FFNN	Ireland	0.4874	0.6226
FFNN	Italy	0.3683	0.4906
FFNN	Korea	0.2558	0.3672
FFNN	Luxembourg	0.5234	1.0439
FFNN	Mexico	0.5705	1.1128
FFNN	Netherlands	0.3782	0.417
FFNN	New Zealand	0.2039	0.2375
FFNN	Norway	0.3498	0.4209
FFNN	Portugal	0.3727	0.4598
FFNN	Spain	0.5622	0.8289
FFNN	Sweden	0.5175	0.5968
FFNN	United Kingdom	0.3975	0.4971
FFNN	United States	2.3094	4.8638
RNN	Australia	0.2984	0.4961
RNN	Austria	0.5486	0.8008
RNN	Belgium	0.3368	0.4263
RNN	Canada	0.9791	2.0102
RNN	Denmark	0.3756	0.494
RNN	Finland	0.3709	0.501
RNN	France	0.2937	0.4244

RNN	Germany	0.1602	0.2022
RNN	Ireland	0.5034	0.6804
RNN	Italy	0.3738	0.4591
RNN	Korea	0.1947	0.2463
RNN	Luxembourg	0.3554	0.5435
RNN	Mexico	0.2565	0.4053
RNN	Netherlands	0.2606	0.3277
RNN	New Zealand	0.1871	0.2573
RNN	Norway	0.3242	0.4007
RNN	Portugal	0.3876	0.5
RNN	Spain	0.4	0.5601
RNN	Sweden	0.3574	0.4518
RNN	United Kingdom	0.189	0.2228
RNN	United States	1.1403	2.6182
ARIMAX	Australia	0.3956	0.5874
ARIMAX	Austria	0.5187	0.7668
ARIMAX	Belgium	0.4132	0.5783
ARIMAX	Canada	1.5059	2.8861
ARIMAX	Denmark	0.4056	0.5629
ARIMAX	Finland	0.4015	0.5316
ARIMAX	France	0.4398	0.7404
ARIMAX	Germany	0.0992	0.127
ARIMAX	Ireland	0.601	0.7666
ARIMAX	Italy	0.5418	0.8384
ARIMAX	Korea	0.2147	0.2602
ARIMAX	Luxembourg	0.3783	0.5821
ARIMAX	Mexico	0.3127	0.4679
ARIMAX	Netherlands	0.3586	0.4449
ARIMAX	New Zealand	0.3435	0.5472
ARIMAX	Norway	0.3577	0.4506
ARIMAX	Portugal	0.5522	0.7246
ARIMAX	Spain	0.7926	1.1053
ARIMAX	Sweden	0.3513	0.4439
ARIMAX	United Kingdom	0.346	0.6374
ARIMAX	United States	1.6873	3.3758

10.3. ARIMAX orders per country

Country	ARIMAX order
Australia	[1, 1, 2]
Austria	[1, 0, 0]
Belgium	[1, 0, 0]
Canada	[0, 1, 1]
Denmark	[1, 0, 0]
Finland	[1, 0, 0]
France	[0, 1, 2]
Germany	[1, 1, 0]
Ireland	[3, 1, 0]
Italy	[2, 1, 0]
Japan	[3, 1, 0]
Korea	[1, 0, 0]
Luxembourg	[0, 1, 1]
Mexico	[1, 0, 0]
Netherlands	[0, 1, 3]
New Zealand	[1, 0, 0]
Norway	[1, 0, 0]
Portugal	[1, 1, 0]
Spain	[2, 1, 0]
Sweden	[2, 0, 0]
United Kingdom	[1, 1, 0]
United States	[2, 0, 0]